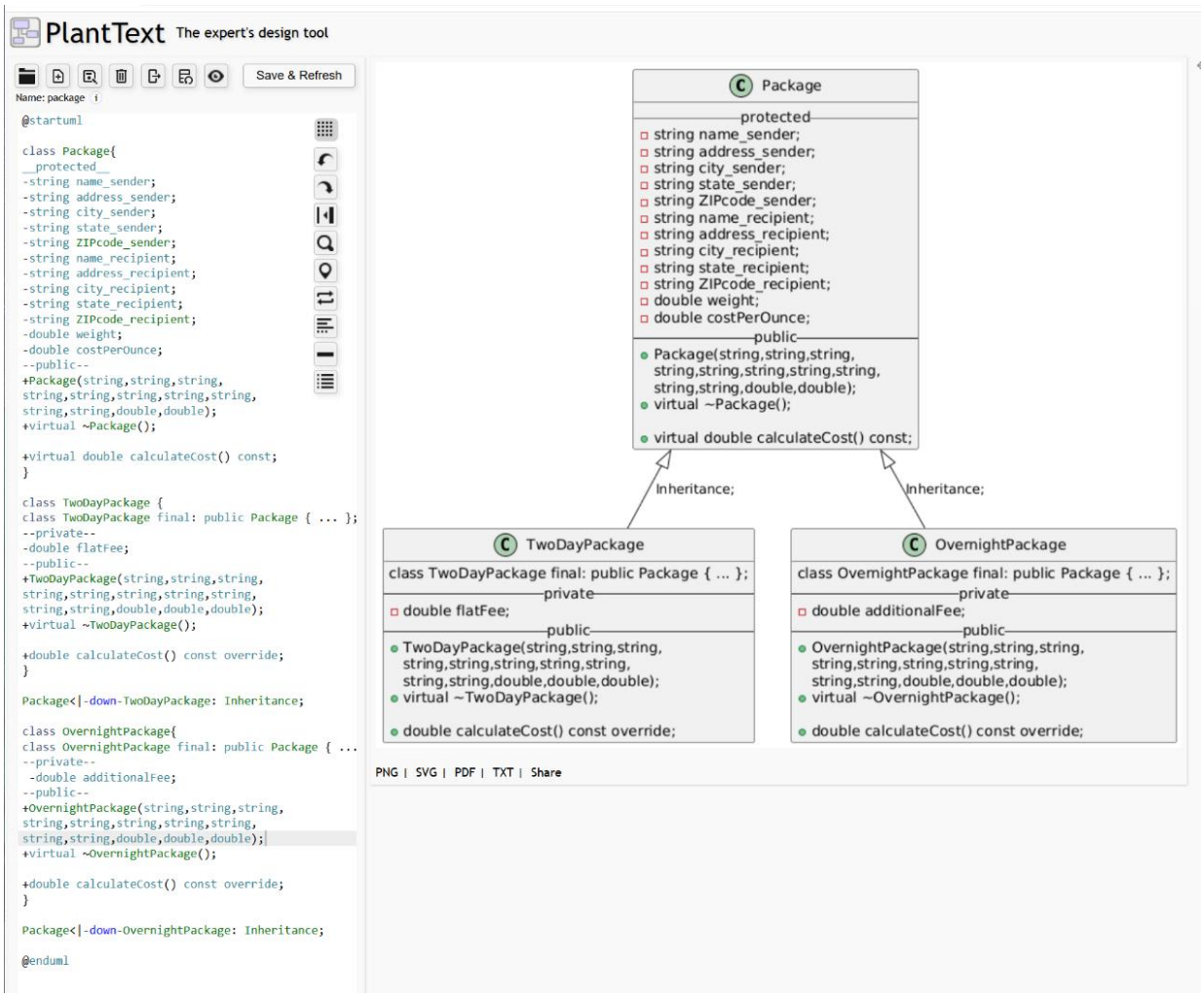
1.4 UML diagram of the inheritance of this package system:



In the first question, Package is the base class, while TwoDayPackage and OvernightPackage are subclasses. The calculateCost function in the base class is declared as virtual to allow overriding in subclasses. Both subclasses can be declared as final since they have no further subclasses under them.

In the second problem, 14 operators were overloaded and tested, and the testMandelbrot function was completed using operator overloading for calculation.